

Comparing Two Speeches

Sydney Wilson - COSC 545

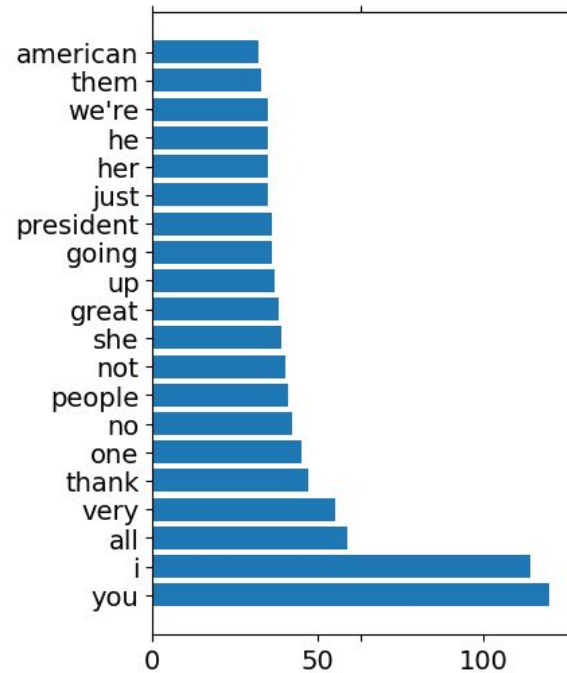
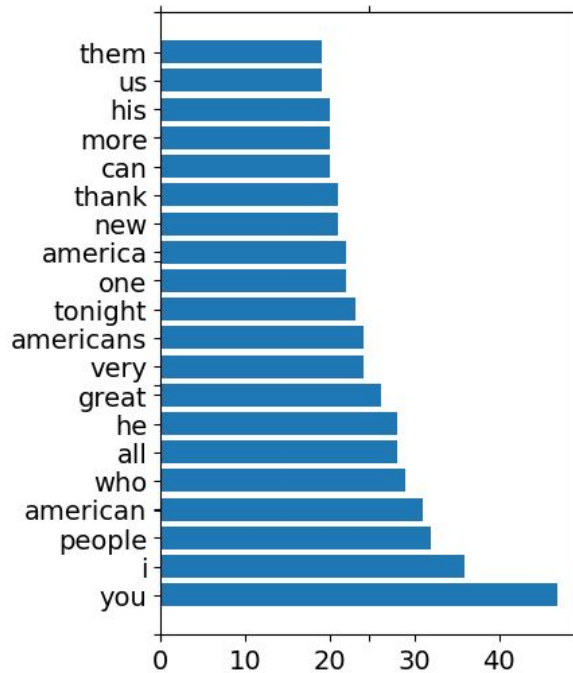
My “Literature”

- I chose to analyze speeches against each other, particularly State of the Union Addresses from our two most recent presidents: Donald Trump and Joe Biden
 - This also fit perfectly as I could compare Trump’s speech from his first and second term for the “same author” comparison.
- My Question(s): Can frequent words would show us the major points or topics of the speech?
- Do the speech's word choices reveal something about how the speaker relates to their audience?

Same Author

Frequent Words Approach

Difference between Donald Trump's State of Address Speech during Each Term



Problem 1

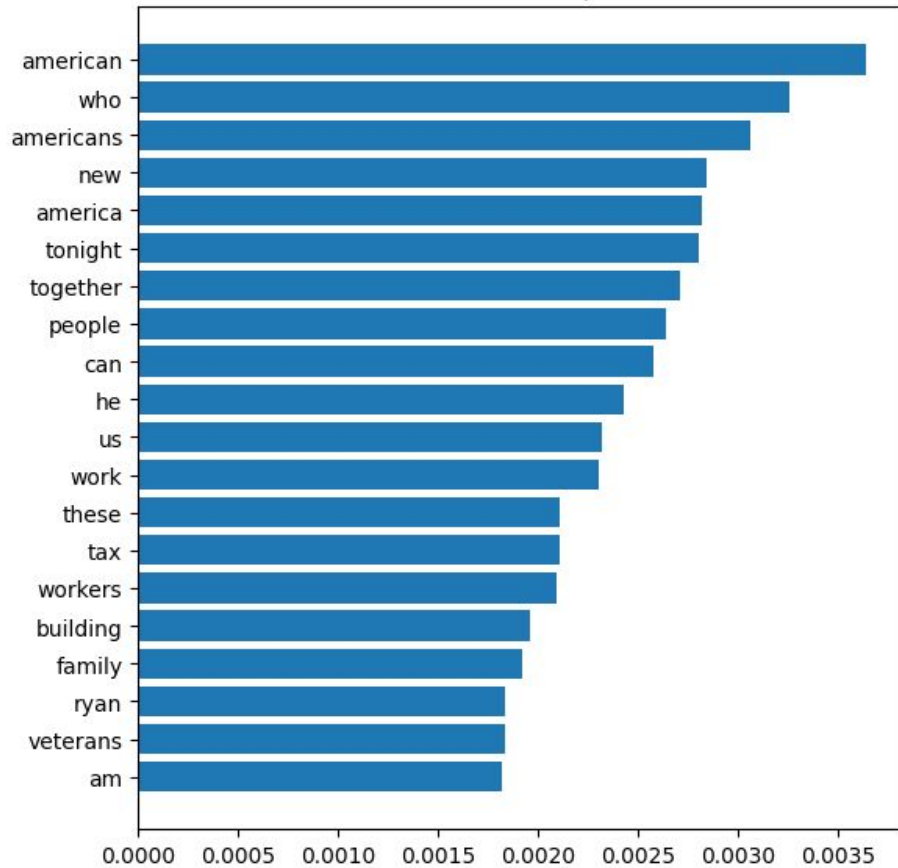
- My first run of the most frequent words returned things that were not very useful: “it”, “has”, etc.
 - I started to resolve this by making a longer stop word list to leave them out of my list, but I eventually settled on the list seen on the last slide.
- I also noticed some “audience reactions” being flagged like “[laughter]” which tell us maybe how the audience was receiving the speech, but do not count as “authored text” which led to me trying to figure out how to filter those things out as well.

New Ideas/Different Analysis

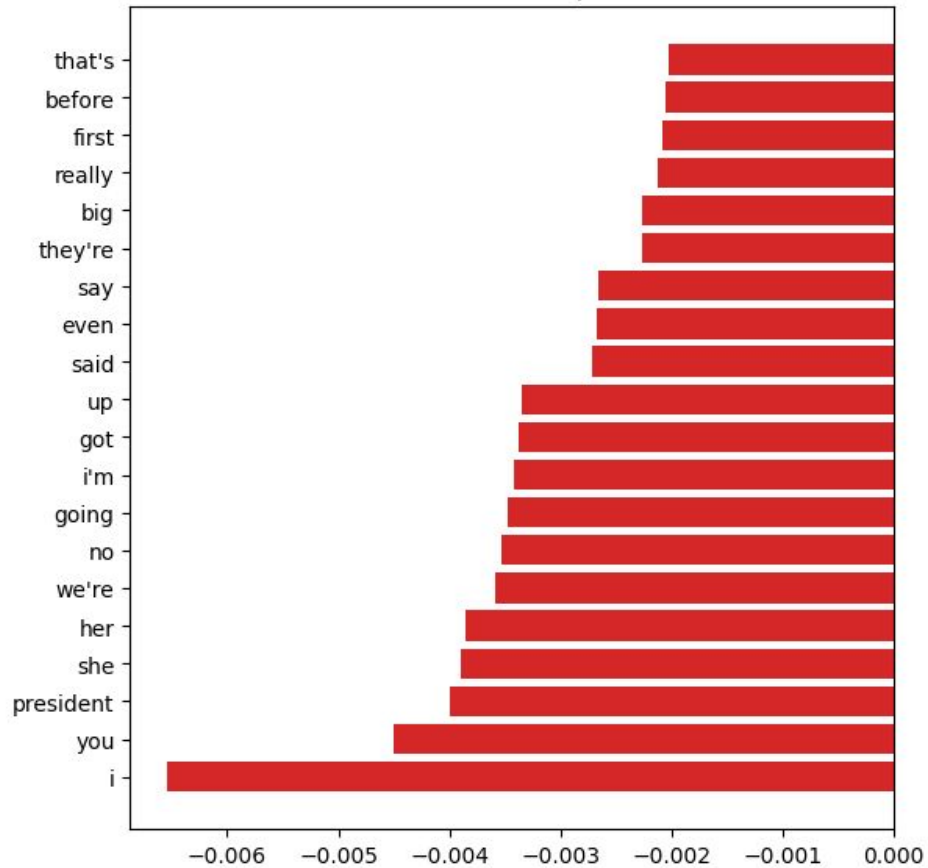
- Even after adjusting stop words, I found that the frequent words still only showed a small portion of the big picture to compare the two speeches.
- I wanted to compare them in a different way like using TF-IDF but I only have two documents, so I settled on relative frequency
 - I wanted to see what words were used a lot in one speech that were rarely if ever used in the other.
- I thought this may highlight a true “difference” in the speeches.

Secondary analysis

More common in Speech 1



More common in Speech 2



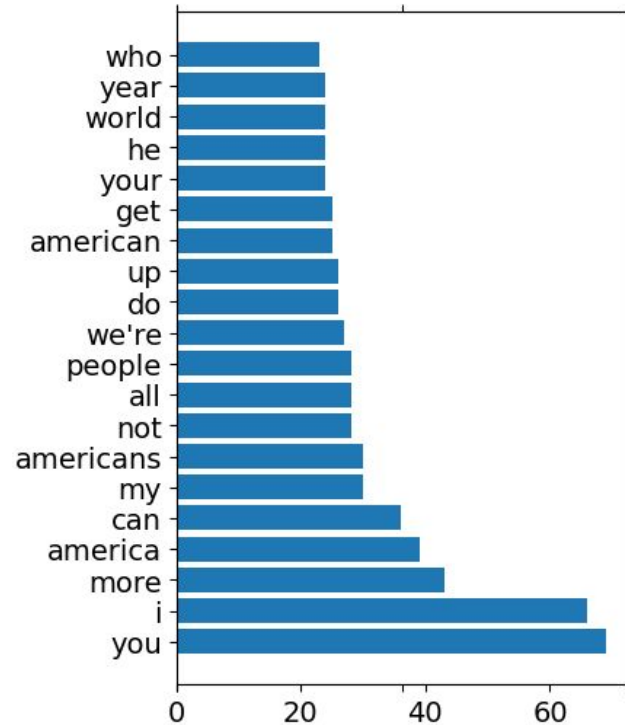
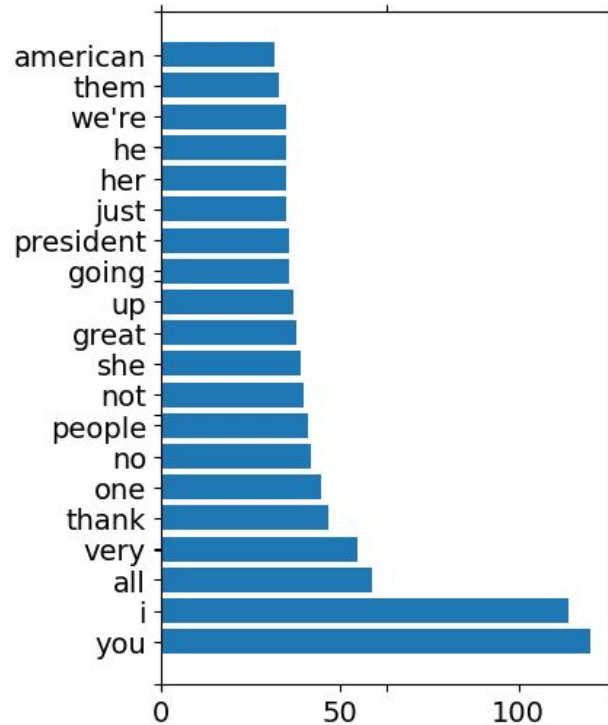
Analysis

- In the first speech, we see words like “veteran, tax, workers, building” which can lead us to see he was speaking directly on issues he planned to tackle: immigration, tax cuts, and military strengthening
 - Ryan is an american police officer who adopted a child in need. Trump used the story to emphasize the goodness in America in order to speak on more “preserving” that goodness by requesting more military funding.
- Speech 2 uses more pronouns and informal, narrative language. Trump spends much of the speech telling stories about individual Americans to make his points, often addressing the audience directly.
 - Sarah Beckstrom is a slain soldier whose story highlights a security/military issue; the U.S. men's hockey team's introduction also uses several of the frequent words

Different Authors

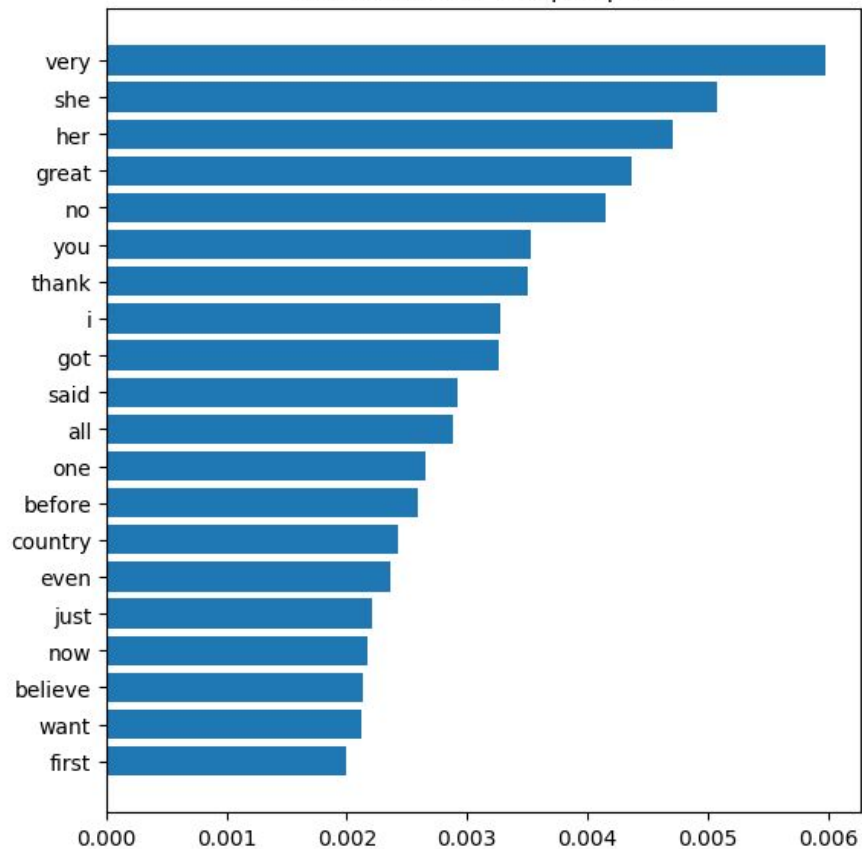
Frequent Words

Difference between Trump`s and Biden`s State of Address Speeches

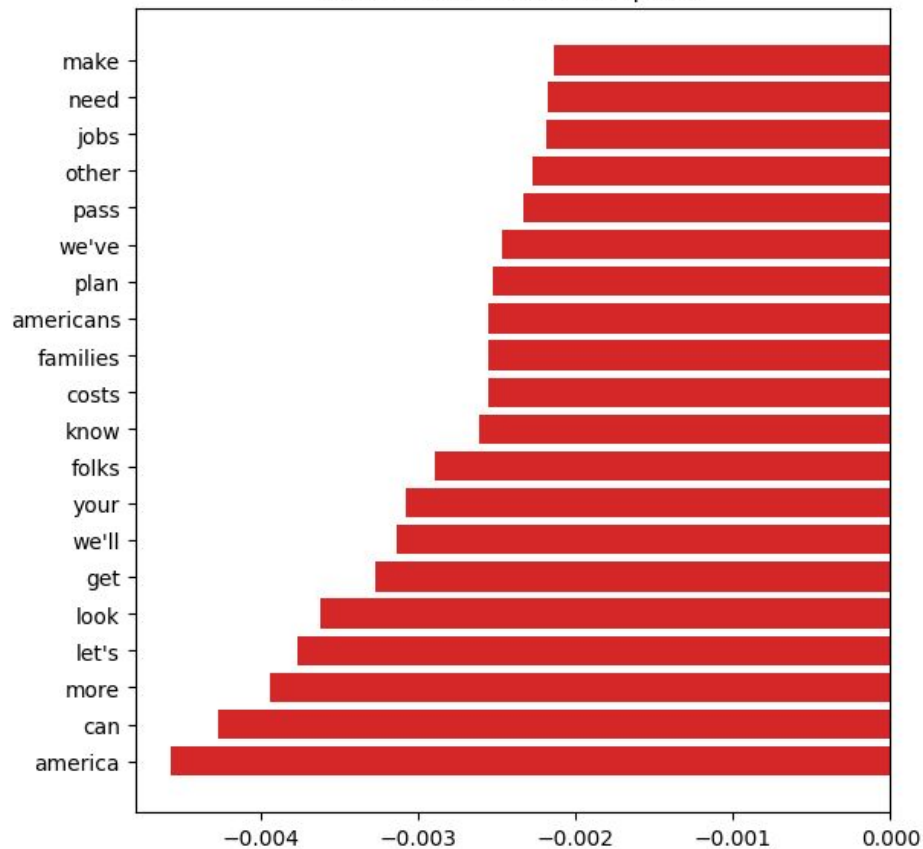


Secondary analysis

More common in Trump's Speech



More common in Biden's Speech



Analysis

- Trump's speech again uses more pronouns and informal, narrative language. Trump spends much of the speech telling stories about individual Americans to make his points, often addressing the audience directly.
 - We also see "believe" and "great" which could be used to position himself or potential plans in a positive light to the audience.
- Biden uses "families, costs, jobs, pass" which most likely refer to future plans and bills he wants to pass. We also see "we" and "let's" to help the audience consider him a teammate.
 - We also see "can" and "more" which can leave the audience with hope for future aspirations mentioned in the speech.

Extra Credit Analysis

- It seemed Biden used a lot of "team-focused" words in his speech, so I checked if I could spot differences along those lines. I decided to test the word "i" across all three speeches. Interestingly, there wasn't much difference between Trump's most recent term SOTU and Biden's ($p = 0.135$), but Trump himself shifted his use of "i" significantly between his own two terms ($p = 0.005$), using it notably more in his second-term address.
- I tried to words that differed between Biden and Trump specifically, so I tested a few more major concepts like immigration, economy, inflation, etc. I found that the word "jobs" was used more by Biden (15/5,194) than Trump (5/7,135), with a p-value of 0.003. I think this again reflects Biden's agenda by promising those watching an increase in jobs.

Summary of Actual Speech Goals

- **Trump, January 30, 2018 SOTU:** Trump touted the economy and the newly passed tax cuts while laying out an agenda on infrastructure, immigration reform, trade, and national security, framing the speech around building a "safe, strong, and proud America."
- **Biden, March 1, 2022 SOTU:** Biden addressed the recent Russian invasion of Ukraine and rallied support for Ukraine, while also pushing a domestic economic agenda focused on jobs, lowering costs for families, and manufacturing, delivered amid high inflation and lingering COVID-19 concerns.
- **Trump, February 24, 2026 SOTU:** Delivered during a partial government shutdown, Trump framed the U.S. as entering a "Golden Age" ahead of its 250th anniversary, citing economic gains, tax cuts, tariff policy, and foreign policy wins, while honoring guests including wounded National Guard members and the U.S. Olympic hockey team.